

# MOHIT RAJPUT

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## SUMMARY

Software Engineer with 2+ years of experience building scalable backend systems and AI-powered applications. Experienced in Python, TypeScript, LLM integration, Retrieval-Augmented Generation (RAG), AI voice agents, Speech-to-Text (STT), Text-to-Speech (TTS), and real-time conversational AI. Strong background in microservices, cloud infrastructure, and designing production-ready AI systems using modern LLM frameworks and vector databases.

## TECHNICAL SKILLS

- **Backend & Languages:** TypeScript, JavaScript (ES6+), Python, FastAPI, NestJS, Node.js, Express.js.
- **AI / Generative AI:** LLMs, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Conversational AI, Voice AI, Context Management, Semantic Search
- **Databases:** DynamoDB (NoSQL), MongoDB, Neo4j (Graph), Elasticsearch/OpenSearch, Firestore, ChromaDB, Vector Databases.
- **Cloud Infrastructure:** AWS (Lambda, CloudFront, Lambda@Edge, S3, EC2), Firebase.
- **DevOps & Tools:** Docker, Jenkins, CI/CD Pipelines, Git, Postman, JIRA, LangChain, LangGraph
- **Architecture:** Microservices, REST API Design, Serverless, Graph Data Modeling, OAuth 2.0, Socket.io.

## EXPERIENCE

### Regenesys Institute of Management Pvt. Ltd

May 2026 – Present

SDE-1

Bengaluru

- \* Designed and developed an AI Voice Calling platform integrating LLMs, Speech-to-Text (Whisper/Deepgram), Text-to-Speech (Deepgram), and telephony services for real-time human-like conversations.
- \* Built Retrieval-Augmented Generation (RAG) pipelines using vector databases and semantic search to enable context-aware responses from enterprise knowledge bases.
- \* Integrated Groq/OpenRouter-hosted LLMs with prompt orchestration and conversation memory for low-latency AI interactions.
- \* Optimized AI response latency through asynchronous processing, streaming pipelines, and efficient context retrieval.

### Tech Matpatra Pvt Ltd

July 2024 – April 2026

SDE-1

Gurugram

- \* Designed and migrated to modular Nest.js microservices with strict TypeScript enforcement, decoupling monoliths and cutting build times by **25%**
- \* Built high-performance AWS Lambda and Lambda@Edge systems, optimizing cold starts and reducing monthly cloud costs by **20%**.
- \* Implemented Elasticsearch/OpenSearch and Neo4j to power low-latency search and graph-based recommendations, delivering up to **3x performance gains**.
- \* Architected DynamoDB single-table designs with GSIs and automated CI/CD using Docker and Jenkins for reliable, scalable deployments.

## PROJECTS

### AI Voice Calling Agent

Python, FastAPI, Groq LLM, Whisper, Deepgram, Twilio, ChromaDB

- \* Designed an end-to-end conversational AI pipeline connecting telephony, streaming STT, LLM inference, RAG retrieval, and TTS for real-time voice interactions.
- \* Integrated LLMs with streaming Speech-to-Text and Text-to-Speech for low-latency conversational experiences.
- \* Implemented conversation memory, prompt management, and interruption handling for human-like interactions.

### Codial : Full Stack Social Media Platform |Source Code

Node.js, Express.s, CookieParser, passport - Oauth, MongoDB, Ejs

- \* Developed a Node.js + Express.js social platform with EJS templating and MongoDB.
- \* Implemented **OAuth** (Google & Local) authentication, **session management**, and core features like posts, comments, and friend requests.
- \* Added **real-time chat** and notifications, enhancing interactivity and user engagement.

## EDUCATION

### Dr. APJ Abdul Kalam Technical University, Lucknow

Bachelor of Technology | Computer Science and Engineering

2020-24

Uttar Pradesh